TECHNICAL IMPLEMENTATION PLAN & DATA REQUIREMENT SPECIFICATION

Automated Vision-Based Milk Pouch & Crate Counting System

Client: Milma (E.R.C.M.P.U. Ltd.), Kottayam Dairy Plant | **Partner:** Goose Industrial Solutions (Vision AI)
Project Scope: High-speed conveyor pouch counting, variant classification & automated crate reconciliation

1. Executive Summary & Project Overview

Following the site survey conducted at the Milma (E.R.C.M.P.U. Ltd.) Kottayam Dairy, Goose Industrial Solutions is pleased to submit this comprehensive Implementation Plan and Data Requirement Specification. The proposed solution provides an automated, non-intrusive Vision-Based Milk Pouch & Crate Counting System designed to operate in real-time directly above the plant's existing crate conveyor line.

The system utilizes industrial global-shutter imaging and deep-learning instance segmentation to count 500 ml pillow pouches packed within blue moulded crates traversing at line speeds of 24 crates/min (~480 pouches/min). Accurate counts will be segregated across the three milk variants (Orange, Violet, Green) to deliver 100% auditable production figures.

Target Facility	Milma (E.R.C.M.P.U. Ltd.), Kottayam Dairy Plant
Throughput Capacity	24 crates/min per line (~280–480 pouches/min) | 1.5 Lakh pouches/day
Product Variants	500 ml LDPE pillow pouches (Orange, Violet, Green) in standard blue crates (20 pouches/crate)
Technology Stack	GigE Global Shutter Vision Camera + Polarized LED Lighting + YOLO Edge AI Inference
Primary Deliverable	Turnkey vision rig, live HMI dashboard, tamper-proof tally records & ERP export

2. Synthesis of Site Survey Findings & Engineering Mitigations

Our engineering design directly addresses the site characteristics captured during the plant survey:

Site Parameter	Survey Observation	Goose Engineering Mitigation
Dim Lighting & Glare	Dim fluorescent tubes; polished stainless steel guards and wet glossy LDPE film create strong specular glare.	High-intensity polarized LED bar illuminator with cross-polarizing lens filter to eliminate specular glare completely.
Abutting Crates	Crates travel abutting with zero inter-crate gap at ~24 crates/min; manual lift-off causes variable dwell time.	Optical crate-edge tracking combined with photoelectric trigger sensor to delineate crate entry and exit boundaries accurately.
Overlapping Pouches	Pouches lie flat and shingled (overlapping in 2 runs, ~20 per crate) with protruding fin seals.	Deep learning instance segmentation (YOLOv8/v11) trained specifically to segment overlapping fin-seal contours.

Washdown Area	Wet dairy washdown environment with cleaning spray and ambient moisture.	IP65/IP67 food-grade stainless steel/anodized camera enclosure and moisture-sealed IP67 cabling.
Mounting Clearance	Conveyor height 900 mm; belt width 450 mm; max overhead clearance to structure ~950 mm.	Custom low-profile overhead C-frame bracket with 8mm/12mm lens tailored for short working distance (WD ~700 mm).

3. Requirements from Milma Kottayam for Model Development

To achieve maximum counting accuracy (target ≥ 99.5%) across all shifts, Goose Industrial Solutions requires specific dataset assets and plant site provisions from Milma Kottayam as outlined below:

3.1 Machine Learning Dataset & Photography Requirements

Machine learning models require varied, real-world examples of pouches in their actual production states. The following photographic dataset is required to train and validate the segmentation neural networks:

Pouch Variant / Condition	Required Volume	Visual Variations to Include
Orange Variant (Standard)	**600 – 800 Photos**	Nominal 20 pouches/crate; varying overlap depths, fin-seal orientation, dry vs. wet pouches.
Violet Variant (Standard)	**600 – 800 Photos**	Full crates on moving conveyor, different print cutoffs, slightly misaligned pouches.
Green Variant (Standard)	**600 – 800 Photos**	Standard packing runs, pouch settling variations, slight crate skewing on guide frames.
Anomalies & Edge Cases	**350 – 500 Photos**	Underfilled crates (16, 17, 18, 19 packets), loose pouches, folded corners, empty crates, worn crates.
High-Speed Video Footages	**20 – 30 Minutes**	Continuous 1080p @ 30/60 fps top-down videos capturing live crate movement, vibration, and manual offload.

■ **DATASET SUMMARY:** Approximately **2,100 to 2,900 still photographs** + **25 minutes of video**. Photos may be captured using standard high-resolution smartphones or DSLRs following the protocol below, or during a dedicated 1-day data collection visit by Goose engineers.

3.2 Photography Capture Protocol

- **Camera Angle:** Direct perpendicular top-down (nadir 90° view) looking straight into the crate, replicating the proposed overhead camera position. Include ±5° to ±10° tilt variations.
- **Working Distance & Framing:** Position camera 700 mm to 900 mm above the conveyor belt so the entire crate (565 mm × 300 mm) plus ~20 mm outer border is framed.
- **Resolution & Sharpness:** Minimum Full HD (1920 × 1080), sharp focus on pouch fin-seal edges, shutter speed ≥ 1/250s if conveyor is moving to eliminate blur.
- **Lighting Conditions:** Capture samples under standard overhead tubes, during morning shift start (07:30), mid-shift, and with wet pouch surfaces.
- **Folder Organization:** Segregate into folders labeled /Orange/, /Violet/, /Green/, and /Edge_Cases/ (with physical count notes).

3.3 Site Infrastructure Provisions Needed from Milma

- **Power Supply:** One single-phase 230V AC (50 Hz) UPS-stabilized power outlet (~500W load) within 3 meters of the inspection station.
- **Network Connectivity:** One Gigabit Ethernet (RJ45) drop or stable plant Wi-Fi access for local dashboard access and data backup.
- **Mounting Clearance:** Support for mounting an overhead rigid C-frame bracket (850–950 mm above conveyor) before the wall aperture / offload hatch.
- **Plant Coordination:** Nomination of an on-site technical contact (Production/Maintenance) to facilitate line access during calibration and UAT.

4. Proposed Vision AI System Architecture

The proposed system is an edge-native industrial vision station. All neural network inference executes locally without reliance on cloud connectivity, ensuring high processing speed (<100 ms per crate) and zero downtime.

Subsystem	Hardware / Software Specification	Function at Kottayam Dairy
Vision Camera	3.2–5.0 MP Global Shutter CMOS (Sony Pregius), GigE Vision, IP67 enclosure	Freezes crate motion at 24 crates/min without blur or rolling distortion.
Lens & Filter	8mm / 12mm Low-Distortion Lens + Polarizing Optical Filter	Frames full 565 mm crate at ~750 mm WD and cuts specular reflections from wet LDPE film.
Active Lighting	High-Intensity Polarized LED Diffuse Illuminator (IP67 food-grade housing)	Provides consistent, shadow-free illumination regardless of ambient factory lighting.
Edge AI Compute	Industrial Fanless Edge AI Box (NVIDIA Jetson Orin / RTX Industrial PC)	Executes deep learning segmentation and tracking in < 80 ms per frame.
Operator HMI	10.1" / 15.6" Touch Screen HMI + Audio-Visual Stack Light	Displays live crate counts, shift totals, and triggers instant alerts for underfilled crates.

5. Project Implementation Plan & Milestones

Goose Industrial Solutions proposes an 8-week structured deployment schedule designed for seamless execution:

Phase	Milestone Focus	Action Items & Deliverables	Timeline
Phase 1	**Data Acquisition & Labelling**	Collect 2,000+ photos/videos across Orange, Violet, Green. Image annotation and polygon segmentation tagging.	**Weeks 1 – 2**
Phase 2	**AI Training & Optimization**	Train YOLO segmentation model. Handle overlapping/shingled pouch edges. TensorRT quantization for edge speed.	**Weeks 3 – 4**
Phase 3	**Hardware Rig Staging**	Fabricate IP65/IP67 enclosure and mounting bracket. Bench test camera, polarized LED light, and Edge AI unit.	**Weeks 4 – 5**
Phase 4	**On-Site Calibration**	Mechanical install at Kottayam line. Calibrate optical trigger under 24 crates/min live motion. Latency benchmarking.	**Weeks 6 – 7**

Phase 5	**UAT & Handover**	Shift 1 (07:30–15:30) validation over 10,000+ crates. Reconcile against physical count. Operator training & sign-off.	**Weeks 8 – 9**

6. Key Performance Indicators & Acceptance Criteria

Performance Metric	Target SLA / KPI	Validation Method
Packet Counting Accuracy	≥ **99.5%**	Audited over 10,000 continuous crates compared against physical manual counts.
Variant Classification	≥ **99.8%**	Automated segregation of Orange vs. Violet vs. Green milk crates.
Inference Latency	**< 100 milliseconds**	Per-crate processing speed (leaves 2.4s buffer at 24 crates/min line speed).
Underfill / Mismatch Alert	≥ **99.0% Detection**	Immediate visual stack light alert and photo archive whenever count != 20.

7. Deliverables & Next Steps

- **Hardware & Rig:** IP65/67 industrial vision camera, polarized LED light, C-arm bracket, and photoelectric trigger.
- **Edge AI Unit & Software:** Pre-loaded industrial edge computer with trained YOLO segmentation and tracking engine.
- **Operator Dashboard:** Touchscreen HMI and local web portal showing live crate counts, shift totals, and alert history.
- **Documentation & Training:** Comprehensive user SOP, maintenance guide, and hands-on plant operator training.

Immediate Next Actions:

1. Review and sign-off on the implementation plan and site infrastructure checklist.
2. Schedule the photo/video dataset collection session at Kottayam plant.
3. Finalize bracket mounting dimensions on the conveyor frame and commence Phase 1.

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